API integration for Imperium Markets

What information do we need the IM platform to pass to our Settlement Coordinator endpoint

Integration Context

- Web 2 - REST API

Endpoint Access Information

- URL - For Accessing the Settlement Coordinator API
- X-Auth-Token // API Key - Header Value for Authentication with API

Endpoints to Use

- Submitting a Deal
- Getting Deal Status
- Notification Service (Maybe Web hook TBD)

Integration Side Note

- Liaise with Michael B to integrate Token Balance view into marketplace UI

Sample JSON Payload - initiate trade transaction

This should be provided to the “Submitting a Deal” endpoint

```
{
  "completionHorizonInSeconds": "<Time limit for Deal Completion (e.g. 3600)",
  "correlationId": "<Unique String Identifier for Deal Flow (e.g.'ABCD-4007')>",
  "participants": {
    "buyer": {
      "participant": "Alias for participant <e.g. blueBank>", // bluebank, redbank, AP+, dev. Would require changes to config files OR a new feature (DB of participants) to add more options here
      "credential": "string",
      "wallet": {
        "offerChain": "hedera",
        "offerAsset": "asset",
        "offerAmount": "Quantity of Currency Offered for Purchase (e.g. 1000000) - Honors EVM Decimals places of currency (e.g. 6)",
        "offerAssetAddress": "EVM Currency Address used for purchase (e.g. 0x...)",
        "deliveryAssetAddress": " EVM Account Address to receive purchased asset (e.g 0x...)"
      }
    },
    "seller": {
      "participant": "Alias for participant <e.g. redBank>",
      "credential": "string",
      "wallet": {
        "bidChain": "hedera",
        "bidAsset": "asset",
        "tokenIssuance": "<Token Issuance (e.g. 'real-time, 'existing')>",  // new
        "tokenType": "<Token Standard (e.g. 'erc-20', 'erc-3643')", // new
        "tokenMetaData": {  // Placeholder for token issuance data
          "term": "4 days",
          "interestRate" : "6%"
```

```
        }
        "bidAmount": "Quantity of Currency to be Offered for Asset Being Sold (e.g 1000000) - Honors EVM Decimal Places for currency (e.g. 6)",
        "bidAssetAddress": "EVM Asset Address being sold (e.g. 0x....)",
        "deliveryAssetAddress": "EVM Account Address to receive payment (e.g. 0x...)"
        "deliveryCurrencyAddress" "EVM Currency Address to receive (e.g. 0x...)"    //new
      }
    }
  },
  "swapConditions": {
    "status": "execution"
  }
}
```

Participant Prerequisites

Custodial Provider Signing Details

Account Context

E.g. Utila

```
  "redBank": {
    "provider": "utila:redBankUT",
    "chains":{
      "network/identifier" : { "vaultAccountId": "<ID Number (e.g. 2)>", "walletId": "<ID Number (e.g. 4)" }
    }
  }
```

E.g. Fireblocks

```
  "redBank": {
    "provider": "fireblocks:redBankFB",
    "chains": {
      "evm:hedera:testnet": { "vaultAccountId": "2", "assetId": "HEDERA_EVM_TEST", "keyType": "ECDSA" }
    }
  }
```

API Authentication

e.g. Utila

```
"utila" : {
  "participantId": {
    "emailAddress": "<Service Accounts Email Address>"
    "publicKey": "<PKI for generating a JWT for API Key>",
    "privateKey": "<PKI for generating a JWT for API Key>",
    "baseUrl": "https://api.utila.io/v1"
  }
}
```

e.g. Fireblocks

```
"fireblocks" : {
  "redBankFB": {
    "apiKey": "<API Key Value from Fireblocks>",
    "privateKey": "<Private Key for SSH Connection>",
    "baseUrl": "https://api.fireblocks.io/v1"
  }
}
```

Reasons for things

1. Custodial wallet information: required to use API co-signing functionality of the custodial wallets, so the human key owner doesn't need to sign every transaction manually

Q&A

1. What transaction types do we support (for pilot demo day)?
   a. Sell an existing (erc-20) RWAT for stablecoins
2. What is not supported
   a. Create an RWAT and sell it for stablecoins a new or existing
   b. RWAT for stablecoins of a different type than the buyer holds (i.e. buy RWAT and swap for different stablecoin in one transaction – we may add this in future)
3. Should we have a database of participants to populate values like buyerID and sellerID
   a. Ideally yes, this would be feature dev for THG. To be included in the next raft of changes
4. Do we need to (for the purposes of pilot) enable token interchange when the seller has a preference to receive currency of a different type than the buyer has?
   a. Yes, ideally, but this is feature dev. We could simulate the same flow by doing two transactions manually (buy RWAT then swap)
5. Should we correlate transactions with our own ID value (e.g. to associate an RWAT purchase txn and a coin swap txn that are part of the same deal), or just use the CorrelationID (a value generated on IM platform side)?
   a. Nice to have, needs development
6. For the purposes of the pilot demo, where do custodial wallet details come from?
   a. AP+ will set up the wallets, and provide details to IM